MATH 231 – Linear Algebra I

The Vector Unit, Part 2 ▪ Orientation, the Dot Product, and Projection

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About this document. This is Part 2 of the vector unit. Section numbering runs continuously through all three parts, so a reference such as §7.2 points into Part 1 and means exactly what it says:

Part	Sections	Subject
1	§1–§9	vectors, notation, addition, scaling, equality
2	§10–§12	orientation, the dot product, projection
3	§13–§15	Cauchy–Schwarz, metrics, norms
4	§16–	the standard basis, span, coordinates

Assumed from Part 1. The Euclidean norm $\|\mathbf{v}\|$ and its properties; $\mathbb{R}^2$ and the signature notation $f : A \to B$; addition, subtraction and scalar multiplication, with the scaling rule $\|\alpha\mathbf{v}\| = |\alpha|\|\mathbf{v}\|$; and the fact that a vector is a displacement, so the same vector may be drawn anywhere.

What you should be able to do afterwards. Say what it means for two vectors to be parallel or orthogonal, in terms of the angle θ between them; derive $\langle \mathbf{u}, \mathbf{v}\rangle = \|\mathbf{u}\|\|\mathbf{v}\|\cos\theta$ from the law of cosines; compute dot products and cosine similarities; test two vectors for orthogonality with a single dot product; normalize a vector and split it into a direction and a magnitude; compute the projection of one vector onto another and the component that goes with it; and split a vector into a piece along $\mathbf{u}$ and a piece orthogonal to $\mathbf{u}$.

Further reading. Boyd & Vandenberghe, *Introduction to Applied Linear Algebra*, Ch. 3 (norm, distance, angle); Strang, *Linear Algebra and Its Applications*, Ch. 1; Axler, *Linear Algebra Done Right*, Ch. 6 (inner products); Trefethen & Bau, *Numerical Linear Algebra*, Lec. 2 (orthogonality).

10. Relative orientation: parallel, orthogonal, and the angle between

Two vectors sitting in $\mathbb{R}^2$ have a relationship beyond their individual magnitudes and directions: how they are oriented *with respect to each other.* Two arrangements are special enough to be named.

10.1. Parallel vectors

> **Parallel (geometric definition).** Two nonzero vectors are **parallel** when, drawn from a common tail, they lie along one and the same straight line. Equivalently: the lines containing their arrows are parallel lines, so the arrows never tilt away from each other — they either point the same way or exactly opposite ways.

§8 already drew this picture without naming it: every scalar multiple of $\mathbf{v}$ lands on the single line through the origin and the head of $\mathbf{v}$. That is precisely the algebraic version of the definition.

> **Parallel (algebraic test).** Nonzero $\mathbf{u}$ and $\mathbf{v}$ are parallel exactly when one is a scalar multiple of the other: $\mathbf{v} = \alpha\mathbf{u}$ for some $\alpha \neq 0$. If $\alpha > 0$ they point the *same* way; if $\alpha < 0$ they point *opposite* ways, and the pair is often called **antiparallel**.

Example 10.1 (Parallel, both signs)**.** $[\,1,\,2\,]$ and $[\,3,\,6\,]$ are parallel and point the same way, since $[3,6] = 3[1,2]$ with $\alpha = 3 > 0$.
$[\,1,\,2\,]$ and $[\,-2,\,-4\,]$ are parallel and antiparallel in direction, since $[-2,-4] = (-2)[1,2]$ with $\alpha = -2 < 0$.
$[\,1,\,2\,]$ and $[\,2,\,1\,]$ are *not* parallel: no single α satisfies both $2 = \alpha \cdot 1$ and $1 = \alpha \cdot 2$.

The zero vector is excluded from the definition on purpose: it has no direction, so asking whether it is parallel to anything is asking a question with no content.

10.2. Orthogonal vectors

> **Orthogonal.** Two nonzero vectors are **orthogonal** when, drawn from a common tail, they meet at a right angle. "Orthogonal" is the word we will use throughout the course; "perpendicular" means the same thing.

The cleanest example is the pair of coordinate directions themselves: $[\,1,\,0\,]$ and $[\,0,\,1\,]$ meet at a right angle at the origin. A less obvious pair is $[\,2,\,1\,]$ and $[\,-1,\,2\,]$: each is the other turned a quarter turn, and both have norm $\sqrt{5}$.

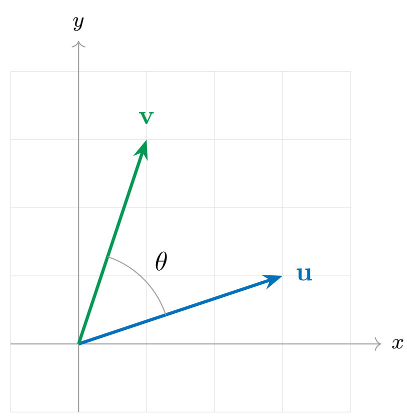
Remark. You already own one test for that last claim: from precalculus, two lines are perpendicular when their slopes multiply to -1. The arrows $[2,1]$ and $[-1,2]$ have slopes $\frac{1}{2}$ and -2, and $\frac{1}{2} \cdot (-2) = -1$. But notice how brittle this test is. It says nothing at all when one of the vectors is vertical — slope undefined — it does not extend to three dimensions or beyond, and it answers only the yes/no question. It cannot tell you how far from perpendicular two vectors are when they are not.

10.3. The angle between two vectors

"Parallel" and "orthogonal" are the two extreme cases of a single underlying quantity, and everything in between has been left unnamed. To measure relative orientation *numerically*, in a way that covers all cases at once, we need the **angle between the vectors**.

Definition 10.1 (The angle θ)**.** Let $\mathbf{u}$ and $\mathbf{v}$ be nonzero vectors in $\mathbb{R}^2$, drawn from a common tail. Write θ for the angle they open up at that shared tail, measured so that

$$\theta \in [\,0,\,\pi\,].$$

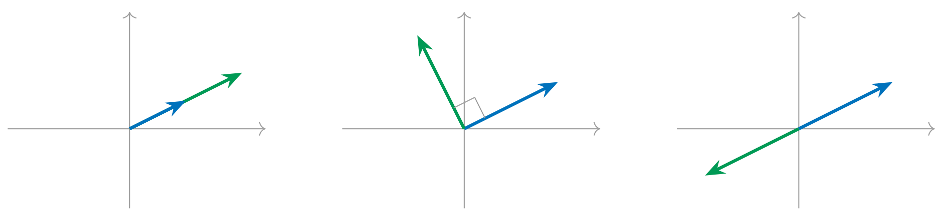


Why that range? Two rays leaving a common point cut the plane into two angles, and those two angles add up to 2π. One of them is the opening *between* the rays; the other is the reflex angle going the long way around outside. Restricting θ to $[0,\pi]$ always selects the first one. Two consequences, both of which we want:

- θ is genuinely the angle *between* the two vectors, never the one sweeping around behind them;
- θ does not depend on which vector you start from — the angle from $\mathbf{u}$ to $\mathbf{v}$ equals the angle from $\mathbf{v}$ to $\mathbf{u}$. It is a property of the *pair*, with no orientation or sign attached.

With θ in hand, both named arrangements become single numbers, and the unnamed middle ground gets a scale:

Angle	**Arrangement**	**Name**
$\theta = 0$	same direction	parallel
$0 < \theta < \frac{\pi}{2}$	opening less than a right angle	acute
$\theta = \frac{\pi}{2}$	right angle	**orthogonal**
$\frac{\pi}{2} < \theta < \pi$	opening more than a right angle	obtuse
$\theta = \pi$	opposite directions	antiparallel



$\theta = 0$: parallel $\theta = \pi/2$: orthogonal $\theta = \pi$: antiparallel

Looking ahead. We can now *say* what relative orientation means, and we could measure θ off the page with a protractor. What we cannot yet do is *compute* it from the components $[u_1, u_2]$ and $[v_1, v_2]$ — nothing we have built so far turns four numbers into an angle. The tool that does it is the *inner product*, and §11 constructs it. It will collapse the whole table above into arithmetic: in particular, testing two vectors for orthogonality will become a single multiplication-and-addition, with no protractor, no slopes, and no exceptions for vertical vectors.

11. The dot product and cosine similarity

Section 10 left us in an awkward position. We can *say* precisely what parallel, antiparallel and orthogonal mean — they are $\theta = 0$, $\theta = \pi$ and $\theta = \pi/2$ — but θ is not something the world hands us. Vectors arrive as pairs of numbers, not as angles, and no amount of staring at $[u_1, u_2]$ and $[v_1, v_2]$ produces a protractor. So:

> Given only the components of two vectors, how do we decide *numerically* whether they are parallel, antiparallel, or orthogonal?

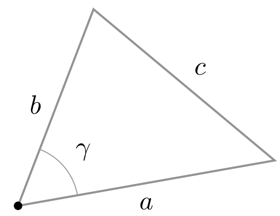
The answer comes from a theorem you already know, applied to a triangle we already know how to build.

11.1. Review: the law of cosines

The law of cosines generalises Pythagoras to triangles that have no right angle.

Law of cosines. In a triangle with two sides of lengths a and b meeting at an angle γ, the third side — the one *opposite* γ — has length c given by

$$c^2 \;=\; a^2 + b^2 - 2ab\cos\gamma.$$



Taking $\gamma = \pi/2$ gives $\cos\gamma = 0$ and the formula collapses to $c^2 = a^2 + b^2$ — Pythagoras is the right-angled special case. The extra term $-2ab\cos\gamma$ is exactly the correction for the angle not being right.

11.2. The triangle built from u, v and u − v

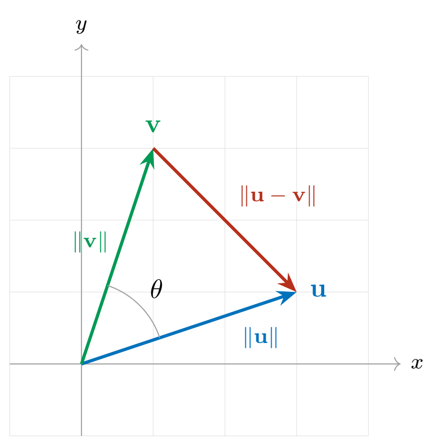
Now build a triangle out of two vectors. Draw $\mathbf{u}$ and $\mathbf{v}$ from a common tail at the origin. Their two heads, together with that shared tail, are three points — a triangle. Two of its sides are the vectors themselves. And the third side, the one joining the two heads, is something we identified back in §7.3: it is $\mathbf{u} - \mathbf{v}$.

So the triangle has side lengths

$$a = \|\mathbf{u}\|, \qquad b = \|\mathbf{v}\|, \qquad c = \|\mathbf{u} - \mathbf{v}\|,$$

and the angle where sides a and b meet — at the common tail — is exactly θ, the angle between $\mathbf{u}$ and $\mathbf{v}$ from §10.3. The side of length c sits opposite it. That is precisely the configuration the law of cosines describes, with $\gamma = \theta$:

$$\|\mathbf{u} - \mathbf{v}\|^2 \;=\; \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\|\mathbf{u}\|\|\mathbf{v}\|\cos\theta. \tag{1}$$



The derivation

Equation (1) is pure geometry: it contains θ but no components. Now compute the same left-hand side *algebraically*, from the components, and compare. With $\mathbf{u} = [u_1, u_2]$ and $\mathbf{v} = [v_1, v_2]$ we have

$\mathbf{u} - \mathbf{v} = [\, u_1 - v_1,\ u_2 - v_2 \,]$, so

$$\begin{aligned}
\|\mathbf{u} - \mathbf{v}\|^2 &= (u_1 - v_1)^2 + (u_2 - v_2)^2 \\
&= \big(u_1^2 - 2u_1v_1 + v_1^2\big) + \big(u_2^2 - 2u_2v_2 + v_2^2\big) \\
&= \underbrace{\big(u_1^2 + u_2^2\big)}_{\|\mathbf{u}\|^2} + \underbrace{\big(v_1^2 + v_2^2\big)}_{\|\mathbf{v}\|^2} - 2\big(u_1v_1 + u_2v_2\big) \\
&= \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\big(u_1v_1 + u_2v_2\big).
\end{aligned}$$

Set this equal to the geometric expression (1):

$$\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\big(u_1v_1 + u_2v_2\big) \;=\; \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\|\mathbf{u}\|\|\mathbf{v}\|\cos\theta.$$

The squared norms cancel from both sides, and dividing by -2 leaves the identity we were after:

$$\boxed{u_1v_1 + u_2v_2 \;=\; \|\mathbf{u}\|\,\|\mathbf{v}\|\cos\theta} \tag{2}$$

where θ is the angle between $\mathbf{u}$ *and* $\mathbf{v}$. That last clause is not decoration: the identity is false for any other angle, and every use we make of (2) depends on θ being the angle between the two vectors in question.

Look at what (2) accomplishes. The left side is built from *components alone* — two multiplications and one addition, no geometry. The right side contains θ. The equation is a bridge between the two, and it is the bridge §10.3 said we were missing.

11.3. The dot product

The left-hand side of (2) is important enough to name.

Definition 11.1 (Dot product / inner product in $\mathbb{R}^2$)**.** For $\mathbf{u} = [u_1, u_2]$ and $\mathbf{v} = [v_1, v_2]$ in $\mathbb{R}^2$, the **dot product** — equivalently the **inner product** — of $\mathbf{u}$ and $\mathbf{v}$ is

$$\langle \mathbf{u}, \mathbf{v} \rangle \;=\; \mathbf{u} \cdot \mathbf{v} \;=\; u_1v_1 + u_2v_2.$$

Both notations are standard and we will use both: $\mathbf{u} \cdot \mathbf{v}$ is common in applied writing, $\langle \mathbf{u}, \mathbf{v} \rangle$ in pure linear algebra. With the name in place, (2) becomes the statement we will quote constantly:

$$\langle \mathbf{u}, \mathbf{v} \rangle \;=\; \|\mathbf{u}\|\,\|\mathbf{v}\|\cos\theta, \qquad \theta = \text{the angle between } \mathbf{u} \text{ and } \mathbf{v}.$$

Remark (A notation trap). Do *not* write $\mathbf{u} \times \mathbf{v}$ for this. That symbol denotes the **cross product**, a completely different operation: it is defined in $\mathbb{R}^3$ rather than $\mathbb{R}^2$, and it returns a *vector* rather than a scalar. Writing $\times$ when you mean $\cdot$ is one of the most common notational errors in a first linear algebra course. Use $\cdot$ or $\langle\,\cdot\,,\cdot\,\rangle$, and reserve $\times$ for the cross product.

Where it sits among our operations

The dot product is our newest **binary operator** — two vectors in, one result out. But notice what makes it structurally new: it is the first binary operator that *leaves* $\mathbb{R}^2$.

$$\langle\,\cdot\,,\cdot\,\rangle \;:\; \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}.$$

Two vectors go in and a *scalar* comes out. Addition and subtraction were binary and stayed inside $\mathbb{R}^2$; the norm left $\mathbb{R}^2$ but was unary. The dot product is the first operation to do both at once.

Operation	Signature	Arity	Returns
Norm $\|\cdot\|$ (§2)	$\mathbb{R}^2 \to \mathbb{R}$	unary	scalar
Addition $+$ (§7)	$\mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}^2$	binary	vector
Subtraction $-$ (§7)	$\mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}^2$	binary	vector
Scalar multiplication (§8)	$\mathbb{R} \times \mathbb{R}^2 \to \mathbb{R}^2$	binary (mixed)	vector
Dot product $\langle \cdot, \cdot \rangle$ (§11)	$\mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R}$	binary	scalar

11.4. Cosine similarity

Solve (2) for the cosine. Since $\theta \in [0, \pi]$ by §10.3, and cosine is decreasing from $\cos 0 = 1$ to $\cos \pi = -1$ across that interval, we know in advance that

$$\cos\theta \in [-1, 1].$$

Dividing (2) by $\|\mathbf{u}\|\|\mathbf{v}\|$ — legal whenever neither vector is the zero vector — gives

$$\frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\|\,\|\mathbf{v}\|} \;=\; \cos\theta.$$

The left-hand side is computable from components. So we may as well promote it to a function in its own right.

Definition 11.2 (Cosine similarity)**.** For nonzero $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$, define

$$\cos(\mathbf{u}, \mathbf{v}) \;:=\; \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\|\,\|\mathbf{v}\|}.$$

This is a function taking *two* vectors in $\mathbb{R}^2$ and returning a real number in $[-1, 1]$. It is called the **cosine similarity** of $\mathbf{u}$ and $\mathbf{v}$, and by the identity above it equals $\cos\theta$, where θ is the angle between $\mathbf{u}$ and $\mathbf{v}$.

The name earns itself twice over. "Cosine," because the output *is* the cosine of the angle between the two vectors. "Similarity," because that output is a score: the closer the two vectors point in the same direction, the closer it is to 1.

Remark (Order does not matter)*.* Cosine similarity is **symmetric**:

$$\cos(\mathbf{u}, \mathbf{v}) = \cos(\mathbf{v}, \mathbf{u}).$$

Both the dot product and the product of norms are unchanged when the two vectors swap places, and §10.3 already established that θ is a property of the *pair* with no preferred starting vector.

A word on the terminology, since the adjectives are easy to swap: a two-argument function is *symmetric* when the *order* of its inputs does not matter, which is what the display above says. *Reflexive* is a different property — it concerns what happens when both inputs are the *same* object. Cosine similarity has a clean property of that kind too, and we compute it in §11.6.

Remark (Scale does not matter either)*.* Because $\cos(\mathbf{u}, \mathbf{v})$ depends only on θ, stretching either vector by a positive factor leaves it unchanged: $\cos(\alpha\mathbf{u}, \mathbf{v}) = \cos(\mathbf{u}, \mathbf{v})$ for every $\alpha > 0$. Cosine similarity measures *direction agreement only* and is blind to magnitude — which is exactly what you want from a similarity score, and exactly what distinguishes it from the dot product itself.

Remark (Similarity kernels)*.* In computational mathematics a function that takes two objects and returns a scalar score of how alike they are is called a **similarity kernel**. Cosine similarity is the first one we meet, and it is by far the most widely used in practice. The objects need not be vectors in $\mathbb{R}^2$ — kernels are built for text documents, images, graphs, and molecules — and we will meet other kernels later in the course.

Reading the three special values

Cosine similarity answers the question this section opened with. Each of the three named arrangements corresponds to one exact value, and the correspondence is made by inverting the cosine on $[0, \pi]$, where arccos is defined and single-valued:

- $\cos(\mathbf{u}, \mathbf{v}) = 1$. Then $\theta = \arccos(1) = 0$: the two vectors open up no angle at all, so they point the same way. $\mathbf{u}$ and $\mathbf{v}$ are **parallel**.
- $\cos(\mathbf{u}, \mathbf{v}) = 0$. Then $\theta = \arccos(0) = \pi/2$: the two vectors form a right angle. $\mathbf{u}$ and $\mathbf{v}$ are **orthogonal**.
- $\cos(\mathbf{u}, \mathbf{v}) = -1$. Then $\theta = \arccos(-1) = \pi$: the opening is a straight line, so the vectors point in exactly opposite directions. $\mathbf{u}$ and $\mathbf{v}$ are **antiparallel**.

Compare this against the table in §10.3. Every row of that table, which we could then only describe, is now a number we can compute from four components.

11.5. When are two vectors numerically similar?

The three special values are the exact cases. What makes cosine similarity useful in practice is everything in between: two vectors are said to be **numerically similar** when $\cos(\mathbf{u}, \mathbf{v}) \approx 1$ — close to pointing the same way, without being exactly parallel.

Example 11.1 (A ladder of similarity)**.** Take $\mathbf{u} = [1, 0]$, so $\|\mathbf{u}\| = 1$, and compare it against several partners.

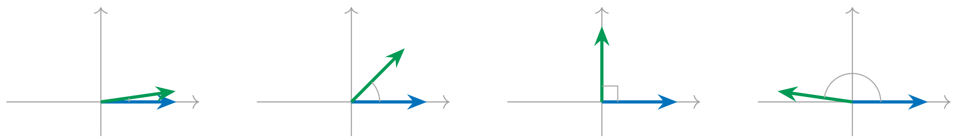
$\mathbf{v}$	$\langle \mathbf{u}, \mathbf{v} \rangle$	$\cos(\mathbf{u}, \mathbf{v})$	**Reading**
$[7, 1]$	7	$7/\sqrt{50} \approx 0.990$	nearly the same direction
$[3, 1]$	3	$3/\sqrt{10} \approx 0.949$	similar
$[1, 1]$	1	$1/\sqrt{2} \approx 0.707$	45° apart
$[0, 1]$	0	0	orthogonal — unrelated
$[-7, 1]$	-7	$-7/\sqrt{50} \approx -0.990$	nearly opposite

The first two rows are what "numerically similar" means. The fourth is the neutral case. The last is as dissimilar as two vectors get.

Example 11.2 (Similar but nowhere near equal)**.** $\mathbf{u} = [3, 4]$ and $\mathbf{v} = [300, 400]$ have $\langle \mathbf{u}, \mathbf{v} \rangle = 900 + 1600 = 2500$, $\|\mathbf{u}\| = 5$ and $\|\mathbf{v}\| = 500$, so

$$\cos(\mathbf{u}, \mathbf{v}) = \frac{2500}{5 \cdot 500} = 1.$$

They are perfectly similar — *parallel* — even though one is a hundred times longer than the other. This is the blindness to magnitude at work: by §9 these are emphatically *not* equal vectors, but their directions agree exactly, and direction is all cosine similarity looks at.



$\cos \approx 0.99$ $\qquad$ $\cos \approx 0.71$ $\qquad$ $\cos = 0$ $\qquad$ $\cos \approx -0.99$

In each panel $\mathbf{u}$ is the horizontal arrow and $\mathbf{v}$ is the other one. Both are drawn at the same length on purpose: cosine similarity cannot see length, only the opening between them.

11.6. A vector against itself

Before computing anything, guess. What should $\cos(\mathbf{u}, \mathbf{u})$ be?

A vector is as similar to itself as anything could possibly be, and it opens no angle at all with itself, so $\theta = 0$ and we expect

$$\cos(\mathbf{u}, \mathbf{u}) \;=\; 1.$$

Now verify it algebraically, straight from the definition. For $\mathbf{u} = [u_1, u_2]$ nonzero,

$$\cos(\mathbf{u}, \mathbf{u}) = \frac{\langle \mathbf{u}, \mathbf{u} \rangle}{\|\mathbf{u}\| \, \|\mathbf{u}\|} = \frac{u_1 u_1 + u_2 u_2}{\|\mathbf{u}\|^2} = \frac{u_1^2 + u_2^2}{\|\mathbf{u}\|^2}.$$

And now look at that numerator. By the definition of the norm in §2, $\|\mathbf{u}\| = \sqrt{u_1^2 + u_2^2}$, so $u_1^2 + u_2^2$ is exactly $\|\mathbf{u}\|^2$. The numerator and denominator are the same number:

$$\cos(\mathbf{u}, \mathbf{u}) = \frac{\|\mathbf{u}\|^2}{\|\mathbf{u}\|^2} = 1,$$

as intuition promised. But the observation we made along the way is worth more than the result:

Identity. For every $\mathbf{u} \in \mathbb{R}^2$,

$$\langle \mathbf{u}, \mathbf{u} \rangle \;=\; \|\mathbf{u}\|^2, \qquad \text{equivalently} \qquad \|\mathbf{u}\| \;=\; \sqrt{\langle \mathbf{u}, \mathbf{u} \rangle}.$$

Dotting a vector with itself returns its squared length. This will do a great deal of work for us in computations and proofs later in the course: it is what lets us convert a statement about lengths into a statement about dot products, and back again, whenever a proof needs one form rather than the other. It is also the first hint of something deeper — that the norm was never independent of the inner product, but a consequence of it.

11.7. The orthogonality test

One special case is used so often that it deserves to be separated from the rest.

Suppose $\mathbf{u}$ and $\mathbf{v}$ are nonzero and orthogonal. Then $\theta = \pi/2$, and $\cos(\pi/2) = 0$, so (2) reads

$$\langle \mathbf{u}, \mathbf{v} \rangle = \|\mathbf{u}\| \, \|\mathbf{v}\| \cos\left(\tfrac{\pi}{2}\right) = \|\mathbf{u}\| \, \|\mathbf{v}\| \cdot 0 = 0.$$

Conversely, if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$ with both vectors nonzero, then $\|\mathbf{u}\|\|\mathbf{v}\| \cos\theta = 0$; the two norms are strictly positive, so $\cos\theta = 0$, and the only angle in $[0, \pi]$ with zero cosine is $\theta = \pi/2$. The implication runs both ways.

Orthogonality test. For nonzero $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$,

$$\mathbf{u} \text{ and } \mathbf{v} \text{ are orthogonal} \qquad \Longleftrightarrow \qquad \langle \mathbf{u}, \mathbf{v} \rangle = 0.$$

This is the payoff promised in §10.3. To test two vectors for orthogonality you do *not* need the angle, the cosine similarity, the norms, or a protractor — and none of the slope-test exceptions apply. Two multiplications, one addition, and check whether the answer is zero.

Example 11.3 (The test in action)**.** Are $[\,2,\ 1\,]$ and $[\,-1,\ 2\,]$ orthogonal? Compute $\langle [2,1], [-1,2] \rangle = 2(-1) + 1(2) = -2 + 2 = 0$. Yes. This confirms by arithmetic what §10.2 argued with slopes — and unlike the slope test, it would work just as well against the vertical vector $[\,0,\ 1\,]$: $\langle [1,0], [0,1] \rangle = 0$.

11.8. Practice: computing cosine similarity

Let

$$\mathbf{u} = [\,1,\ 2\,], \qquad \mathbf{v} = [\,2,\ 4\,], \qquad \mathbf{w} = [\,-2,\ 1\,], \qquad \mathbf{z} = [\,-3,\ -6\,].$$

Compute $\cos(\mathbf{u}, \mathbf{v})$, $\cos(\mathbf{u}, \mathbf{w})$ and $\cos(\mathbf{u}, \mathbf{z})$, and in each case name the arrangement. Then compute $\cos(\mathbf{v}, \mathbf{w})$.

The norms you will need: $\|\mathbf{u}\| = \sqrt{5}$, $\|\mathbf{v}\| = \sqrt{20} = 2\sqrt{5}$, $\|\mathbf{w}\| = \sqrt{5}$, $\|\mathbf{z}\| = \sqrt{45} = 3\sqrt{5}$.

Answer key

Pair	**Dot product**	**Cosine similarity**	**Arrangement**
$\mathbf{u}, \mathbf{v}$	$1(2) + 2(4) = 10$	$\dfrac{10}{\sqrt{5} \cdot 2\sqrt{5}} = \dfrac{10}{10} = 1$	parallel
$\mathbf{u}, \mathbf{w}$	$1(-2) + 2(1) = 0$	$\dfrac{0}{\sqrt{5} \cdot \sqrt{5}} = 0$	orthogonal
$\mathbf{u}, \mathbf{z}$	$1(-3) + 2(-6) = -15$	$\dfrac{-15}{\sqrt{5} \cdot 3\sqrt{5}} = \dfrac{-15}{15} = -1$	antiparallel
$\mathbf{v}, \mathbf{w}$	$2(-2) + 4(1) = 0$	$\dfrac{0}{2\sqrt{5} \cdot \sqrt{5}} = 0$	orthogonal

Remark (Why four vectors and not three?)*.* Because three cannot exhibit all three arrangements. To realise "parallel" or "antiparallel" at all, two of the three vectors must lie on a common line through the origin — and once they do, a third vector's relation to one of them determines its relation to the other. That is exactly what the last row above illustrates: $\mathbf{v} = 2\mathbf{u}$ lies on $\mathbf{u}$'s line, so $\mathbf{w}$, being orthogonal to $\mathbf{u}$, has no choice but to be orthogonal to $\mathbf{v}$ as well. Two of the three pairs therefore always carry the same label, and {parallel, antiparallel, orthogonal} can never appear once each. A fourth vector is the minimum needed.

12. Unit vectors and projection

Since §1 we have said that a vector carries two attributes, a magnitude and a direction. So far we have only ever handled them together. This section pulls them apart — a **direction–magnitude decomposition**, if you like — and then uses the pieces to answer a genuinely new question: how much of one vector points along another?

12.1. Normalization and unit vectors

To isolate the direction of a vector, throw away its magnitude by dividing it out.

Definition 12.1 (Normalization)**.** A nonzero vector $\mathbf{u} \in \mathbb{R}^2$ is **normalized** by dividing it by its own magnitude. We write

$$\hat{\mathbf{u}} := \frac{\mathbf{u}}{\|\mathbf{u}\|} = \frac{1}{\|\mathbf{u}\|}\,\mathbf{u},$$

read "**u**-hat." The result has length 1, and a vector of length 1 is called a **unit vector**; we also say it has **unit length**.

Why does normalizing produce length 1? Because dividing by $\|\mathbf{u}\|$ is scalar multiplication by $\alpha = 1/\|\mathbf{u}\|$, and §8 told us exactly what scaling does to a norm:

$$\|\hat{\mathbf{u}}\| = \left\| \tfrac{1}{\|\mathbf{u}\|}\,\mathbf{u} \right\| = \left| \tfrac{1}{\|\mathbf{u}\|} \right| \|\mathbf{u}\| = \frac{\|\mathbf{u}\|}{\|\mathbf{u}\|} = 1.$$

And since $1/\|\mathbf{u}\| > 0$, the sign rule of §8.1 says the direction is untouched. Normalizing therefore does precisely one thing: it discards the magnitude and keeps the direction. That is the whole point — we normalize when we want to talk about a vector's direction *only.*

Example 12.1 (Normalizing)**.** For $\mathbf{u} = [\,3,\,4\,]$ we have $\|\mathbf{u}\| = 5$, so

$$\hat{\mathbf{u}} = \frac{1}{5}[\,3,\,4\,] = [\,0.6,\,0.8\,],$$

and indeed $\|\hat{\mathbf{u}}\| = \sqrt{0.36 + 0.64} = \sqrt{1} = 1$. The vector $\hat{\mathbf{u}}$ points the same way as $\mathbf{u}$ but is five times shorter.

The zero vector cannot be normalized — the division is undefined, which is the arithmetic shadow of the fact recorded in §8 that $[\,0,\,0\,]$ has no direction to extract.

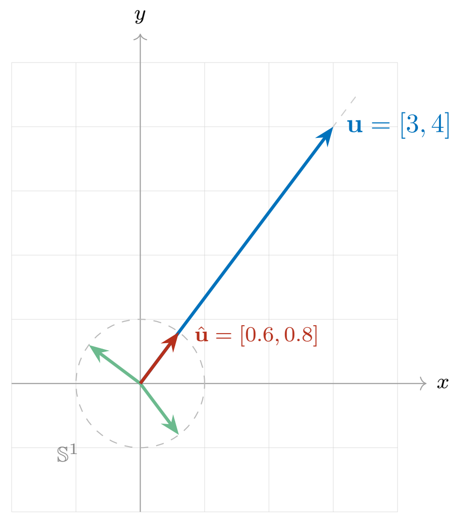
Where all the unit vectors live

A unit vector satisfies $\|\mathbf{u}\| = 1$, that is $\sqrt{u_1^2 + u_2^2} = 1$, or equivalently

$$u_1^2 + u_2^2 = 1.$$

That is the equation of a circle of radius 1 centred at the origin. So:

> Under the Euclidean norm, the heads of *all* unit vectors lie on a single circle of radius 1 about the origin — and every point of that circle is the head of a unit vector.



Looking ahead. The circle is an artefact of the *Euclidean* norm, not of normalization itself. In §14 we will define other norms on $\mathbb{R}^2$, and each one has its own set of "vectors of length 1" — which turns out to be a different shape in the plane for each norm. The circle is only the first of a family of pictures.

The set-theoretic statement

That circle has a name, and it lets us say "$\mathbf{u}$ has unit length" in the notation of §6.

Definition 12.2 (The 1-sphere)**.**

$$\mathbb{S}^1 \;:=\; \big\{\, \mathbf{u} \in \mathbb{R}^2 \;:\; \|\mathbf{u}\| = 1 \,\big\}.$$

Writing $\mathbf{u} \in \mathbb{S}^1$ is a formal way of saying that $\mathbf{u}$ is a unit vector.

Remark. The superscript counts the dimension of the *sphere itself*, not of the space it sits in. The 1-sphere $\mathbb{S}^1$ is a circle: a one-dimensional curve living in the two-dimensional plane — you need only one number, an angle, to say where you are on it. The 2-sphere $\mathbb{S}^2$ is the ordinary sphere: a two-dimensional surface living in three-dimensional space. So a circle is a "one-dimensional sphere," and the object you would casually call "a sphere" is $\mathbb{S}^2$.

12.2. The direction–magnitude decomposition

Now put the two attributes back together deliberately. Multiply and divide any nonzero $\mathbf{u}$ by its own norm:

$$\mathbf{u} \;=\; \underbrace{\frac{\mathbf{u}}{\|\mathbf{u}\|}}_{\text{direction}} \;\cdot\; \underbrace{\|\mathbf{u}\|}_{\text{magnitude}} \;=\; \hat{\mathbf{u}}\,\|\mathbf{u}\|.$$

The first factor is a unit vector carrying the direction and nothing else; the second is a positive scalar carrying the magnitude and nothing else. Every nonzero vector splits this way, and the split is unique:

> **Direction–magnitude decomposition.** For every nonzero $\mathbf{u} \in \mathbb{R}^2$ there are a unit vector $\hat{\mathbf{u}} \in \mathbb{S}^1$ and a scalar $\alpha \in \mathbb{R}$ with $\alpha > 0$ such that
>
> $$\mathbf{u} \;=\; \alpha\,\hat{\mathbf{u}},$$

and they are uniquely determined: $\alpha = \|\mathbf{u}\|$ and $\hat{\mathbf{u}} = \mathbf{u}/\|\mathbf{u}\|$.

Remark. Two fine points, both worth stating once. First, the restriction $\alpha > 0$ is what buys uniqueness: if negative scalars were allowed we could also write $\mathbf{u} = (-\|\mathbf{u}\|)(-\hat{\mathbf{u}})$, since $-\hat{\mathbf{u}}$ is a unit vector too by §8.2. Second, the zero vector is genuinely excluded — it has no direction, so there is no unit vector to factor out of it.

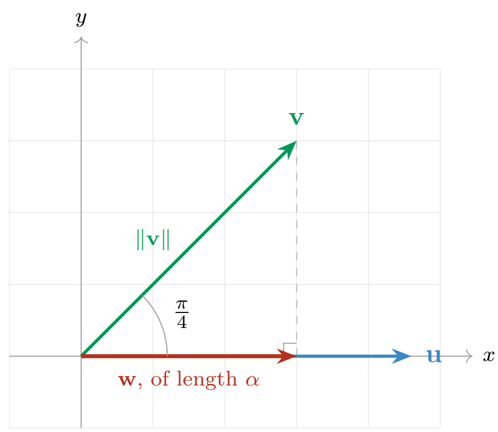
From here on, whenever we want "a vector pointing that way, with that length," we will build it as (scalar) $\times$ (unit vector). The next subsection is the first place we need to.

12.3. The shadow of one vector on another

Here is the question. Let $\mathbf{u}$ be a vector lying along the x-axis, and let $\mathbf{v}$ be a vector making an angle of 45° — that is, $\pi/4$ — with $\mathbf{u}$.

> How much of $\mathbf{v}$ points in the direction of $\mathbf{u}$?

Put physically: if light shone straight down onto the line carrying $\mathbf{u}$, what *shadow* would $\mathbf{v}$ cast on it? Call that shadow $\mathbf{w}$. To find it, drop a line from the head of $\mathbf{v}$ perpendicular to $\mathbf{u}$; where it lands is the head of the shadow.



Deriving the shadow

Step 1 — the direction. The shadow lies along $\mathbf{u}$, so it points in the direction $\hat{\mathbf{u}} = \mathbf{u}/\|\mathbf{u}\|$. By the decomposition of §12.2 we may therefore write it as

$$\mathbf{w} \;=\; \frac{\mathbf{u}}{\|\mathbf{u}\|} \cdot \alpha,$$

where α is the length of the shadow. This is exactly why we built the decomposition first: we know the direction, we do not yet know the magnitude, so we write the vector in the form that separates them.

Step 2 — the length, by trigonometry. The perpendicular we dropped creates a right triangle. Its hypotenuse is $\mathbf{v}$, of length $\|\mathbf{v}\|$; the leg along $\mathbf{u}$ is the shadow, of length α; and the angle between them is $\pi/4$. Cosine is adjacent over hypotenuse, so

$$\cos\frac{\pi}{4} = \frac{\alpha}{\|\mathbf{v}\|}, \qquad \text{hence} \qquad \alpha = \|\mathbf{v}\| \cos\frac{\pi}{4}.$$

Step 3 — replace the angle by the cosine similarity. Here $\pi/4$ is the angle between $\mathbf{u}$ and $\mathbf{v}$, so by §11.4 it is exactly what $\cos(\mathbf{u}, \mathbf{v})$ computes: $\cos(\mathbf{u}, \mathbf{v}) = \cos(\pi/4)$. Substituting,

$$\mathbf{w} \;=\; \frac{\mathbf{u}}{\|\mathbf{u}\|} \cdot \cos(\mathbf{u}, \mathbf{v}) \cdot \|\mathbf{v}\|.$$

Step 4 — unfold the cosine similarity. Now use the definition $\cos(\mathbf{u}, \mathbf{v}) = \langle \mathbf{u}, \mathbf{v} \rangle / (\|\mathbf{u}\| \|\mathbf{v}\|)$:

$$\mathbf{w} = \frac{\mathbf{u}}{\|\mathbf{u}\|} \cdot \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\|\,\|\mathbf{v}\|} \cdot \|\mathbf{v}\| = \frac{\mathbf{u}}{\|\mathbf{u}\|} \cdot \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\|} = \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\|^2}\,\mathbf{u}.$$

The $\|\mathbf{v}\|$ cancels, and with it the last trace of geometry: no angle, no protractor, no triangle. The shadow is computable from the components of $\mathbf{u}$ and $\mathbf{v}$ alone.

Written back in direction–magnitude form, the same vector reads

$$\mathbf{w} \;=\; \underbrace{\frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\|}}_{\text{magnitude}} \;\cdot\; \underbrace{\frac{\mathbf{u}}{\|\mathbf{u}\|}}_{\text{direction}} \;.$$

12.4. Projection and component

The shadow vector has a formal name.

Definition 12.3 (Projection and component)**.** For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^2$ with $\mathbf{u} \neq \mathbf{0}$, the **projection of $\mathbf{v}$ onto $\mathbf{u}$** is

$$\operatorname{proj}_{\mathbf{u}}(\mathbf{v}) \;:=\; \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\|} \cdot \frac{\mathbf{u}}{\|\mathbf{u}\|} \;=\; \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\|^2}\,\mathbf{u},$$

and the scalar part of it — the length of the shadow — is the **component of $\mathbf{v}$ along $\mathbf{u}$**,

$$\operatorname{comp}_{\mathbf{u}}(\mathbf{v}) \;:=\; \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\|}.$$

The two are related by exactly the decomposition of §12.2 — component is the magnitude, $\hat{\mathbf{u}}$ is the direction:

$$\operatorname{proj}_{\mathbf{u}}(\mathbf{v}) \;=\; \operatorname{comp}_{\mathbf{u}}(\mathbf{v}) \cdot \hat{\mathbf{u}}.$$

The projection is a *vector*; the component is a *scalar*. Keeping that straight is most of the battle.

Remark (The component is signed). Although we called α “the length of the shadow,” $\operatorname{comp}_{\mathbf{u}}(\mathbf{v})$ can be negative, because $\langle \mathbf{u}, \mathbf{v} \rangle$ can be. Reading it off the table in §11.4: the component is positive when θ is acute (so $\mathbf{v}$ leans toward $\mathbf{u}$), zero when the two are orthogonal (no shadow at all), and negative when θ is obtuse (so $\mathbf{v}$ leans away, and the shadow falls on the far side of the origin, pointing against $\mathbf{u}$). It is a *signed* length.

Remark (Order matters). The subscript names the vector you are projecting *onto*, and swapping the two arguments gives a different answer: $\operatorname{proj}_{\mathbf{u}}(\mathbf{v}) \neq \operatorname{proj}_{\mathbf{v}}(\mathbf{u})$ in general. The numerator $\langle \mathbf{u}, \mathbf{v} \rangle$ is symmetric, but the $\|\mathbf{u}\|^2$ in the denominator and the $\mathbf{u}$ outside it are not. Problems P1 and P2 below are the same two vectors in the two orders; compare the answers.

Example 12.2 (The motivating case, both ways)**.** Take $\mathbf{u} = [\,2,\ 0\,]$ along the x-axis and $\mathbf{v} = [\,3,\ 3\,]$, which does make an angle of $\pi/4$ with it. By the formula: $\langle \mathbf{u}, \mathbf{v} \rangle = 2(3) + 0(3) = 6$ and $\|\mathbf{u}\|^2 = 4$, so

$$\operatorname{comp}_{\mathbf{u}}(\mathbf{v}) = \frac{6}{2} = 3, \qquad \operatorname{proj}_{\mathbf{u}}(\mathbf{v}) = \frac{6}{4}[\,2,\ 0\,] = \tfrac{3}{2}[\,2,\ 0\,] = [\,3,\ 0\,].$$

Now check it against the trigonometry of Step 2, which never mentioned components: $\|\mathbf{v}\| = \sqrt{18} = 3\sqrt{2}$ and $\cos(\pi/4) = \sqrt{2}/2$, so

$$\alpha = \|\mathbf{v}\| \cos\frac{\pi}{4} = 3\sqrt{2}\cdot\frac{\sqrt{2}}{2} = \frac{3\cdot 2}{2} = 3.$$

The two routes agree. And the answer is the one the picture demands: the shadow of $[3, 3]$ on the x-axis reaches out to $x = 3$.

12.5. Practice: projections

For each pair, compute $\mathrm{comp}_{\mathbf{u}}(\mathbf{v})$ and $\mathrm{proj}_{\mathbf{u}}(\mathbf{v})$.

P1. $\mathbf{u} = [\,1,\,0\,]$, $\quad \mathbf{v} = [\,3,\,4\,]$.

P2. $\mathbf{u} = [\,3,\,4\,]$, $\quad \mathbf{v} = [\,1,\,0\,]$.

P3. $\mathbf{u} = [\,1,\,2\,]$, $\quad \mathbf{v} = [\,4,\,-3\,]$.

Answer key

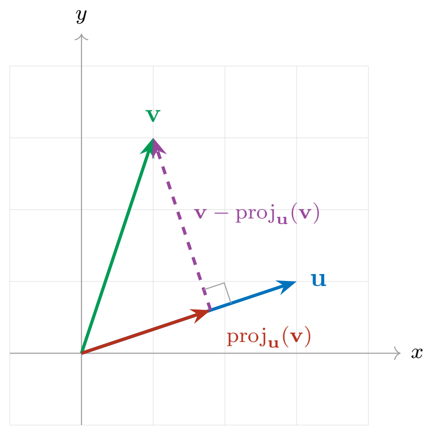
#	$\langle \mathbf{u}, \mathbf{v}\rangle$	$\|\mathbf{u}\|^2$	$\mathrm{comp}_{\mathbf{u}}(\mathbf{v})$	$\mathrm{proj}_{\mathbf{u}}(\mathbf{v})$
P1	$1(3) + 0(4) = 3$	1	$\frac{3}{1} = 3$	$\frac{3}{1}[1, 0] = [\,3,\,0\,]$
P2	$3(1) + 4(0) = 3$	25	$\frac{3}{5}$	$\frac{3}{25}[3, 4] = [\,\frac{9}{25},\,\frac{12}{25}\,]$
P3	$1(4) + 2(-3) = -2$	5	$-\frac{2}{\sqrt{5}}$	$-\frac{2}{5}[1, 2] = [\,-\frac{2}{5},\,-\frac{4}{5}\,]$

Three things to take from these. **P1**: projecting onto $[\,1,\,0\,]$ simply keeps the first coordinate and discards the second, which is what projecting onto the x-axis ought to do — a good sanity check that the formula is not doing anything exotic. **P2**: the same two vectors in the opposite order give a completely different answer, even though the dot product in the numerator is identical. **P3**: the component is negative, so the projection points *against* $\mathbf{u}$; here $\mathbf{v}$ leans away from $\mathbf{u}$, at an obtuse angle to it.

12.6. The orthogonal piece

Return to the picture. We dropped a perpendicular from the head of $\mathbf{v}$ down to $\mathbf{u}$ and used its foot to locate the shadow. But that perpendicular segment is itself a vector, and it is worth naming: it runs from the head of $\mathrm{proj}_{\mathbf{u}}(\mathbf{v})$ to the head of $\mathbf{v}$, so by the between-the-heads reading of §7.3 it is

$$\mathbf{v} - \mathrm{proj}_{\mathbf{u}}(\mathbf{v}).$$



Notice first that the two pieces reassemble $\mathbf{v}$, head to tail:

$$\mathbf{v} \;=\; \underbrace{\mathrm{proj}_{\mathbf{u}}(\mathbf{v})}_{\text{the part along } \mathbf{u}} \;+\; \underbrace{\big(\mathbf{v} - \mathrm{proj}_{\mathbf{u}}(\mathbf{v})\big)}_{\text{the part orthogonal to } \mathbf{u}} .$$

Subtracting off $\mathrm{proj}_{\mathbf{u}}(\mathbf{v})$ removes from $\mathbf{v}$ exactly its directional contribution along $\mathbf{u}$ — everything that pointed that way is gone, and what remains cannot point along $\mathbf{u}$ at all. The claim implicit in the labelling is that the remainder is genuinely orthogonal to $\mathbf{u}$, and it is short work to check.

Verification

Write $c = \langle \mathbf{u}, \mathbf{v}\rangle / \|\mathbf{u}\|^2$, so that $\mathrm{proj}_{\mathbf{u}}(\mathbf{v}) = c\,\mathbf{u}$ and $\mathbf{v} - \mathrm{proj}_{\mathbf{u}}(\mathbf{v}) = [\,v_1 - cu_1,\; v_2 - cu_2\,]$. Dot it against $\mathbf{u}$:

$$\begin{aligned}
\big\langle \mathbf{u},\; \mathbf{v} - \mathrm{proj}_{\mathbf{u}}(\mathbf{v})\big\rangle &= u_1(v_1 - cu_1) + u_2(v_2 - cu_2) \\
&= \big(u_1v_1 + u_2v_2\big) - c\big(u_1^2 + u_2^2\big) \\
&= \langle \mathbf{u}, \mathbf{v}\rangle - c\,\|\mathbf{u}\|^2 \\
&= \langle \mathbf{u}, \mathbf{v}\rangle - \frac{\langle \mathbf{u}, \mathbf{v}\rangle}{\|\mathbf{u}\|^2} \cdot \|\mathbf{u}\|^2 \;=\; 0.
\end{aligned}$$

By the orthogonality test of §11.7, a dot product of zero means orthogonal. Note which identity did the work in the third line: $u_1^2 + u_2^2 = \|\mathbf{u}\|^2$, the $\langle \mathbf{u}, \mathbf{u}\rangle = \|\mathbf{u}\|^2$ of §11.6. That is the promised payoff — the identity is what lets $\|\mathbf{u}\|^2$ cancel against the denominator of c.

Looking ahead. Splitting a vector into a piece along $\mathbf{u}$ and a piece orthogonal to $\mathbf{u}$ is not a one-off trick; it is the engine of a great deal of computational linear algebra. Run it repeatedly on a list of vectors and you manufacture a whole orthogonal family out of an arbitrary one — which is the idea behind the Gram–Schmidt procedure and, through it, the QR factorization and the numerical algorithms built on top of them. We will meet all of these later; for now, notice only how little machinery it took: a dot product, a norm, and a subtraction.